

Dual Gagliardo–Nirenberg–Sobolev embedding

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July 7, 2021

Proposition 1. *Let $g \in L_d(\mathbb{R}^d)$. There exist bounded functions $g_1, g_2, \dots, g_d$ such that*

$$\sum_{j=1}^d \partial_j g_j = g \quad \text{and} \quad \sum_{j=1}^d \|g_j\|_{L_\infty} \lesssim \|g\|_{L_d}. \quad (1)$$

The sign “ $\lesssim$ ” means the inequality holds with a certain uniform constant. One may derive Proposition 1 from the Gagliardo–Nirenberg–Sobolev embedding $\dot{W}_1^1(\mathbb{R}^d) \hookrightarrow L_{d/(d-1)}$ using duality. The purpose of this short note is to present an explicit construction of the solutions to (1). A much more powerful construction (with additional control of the $\dot{W}_d^1$ norms of the g_j in (1)) is presented in [1]. Nevertheless, our reasoning still might be interesting since it is much simpler. In fact, it somehow dualizes the original proof of Gagliardo and Nirenberg. We consider only real-valued functions in this note (the reasoning works in the complex-valued case as well).

Proof. We will induct on the dimension d . Let us consider the case $d = 2$ as a base for our induction. Formally, we may have started with $d = 1$. The case $d = 2$ will help to understand the induction step better.

Case $d = 2$. Set $\partial_1 g_1 = g\psi$, where ψ is a function to be chosen later. What conditions should we impose on ψ in order the function g_1 to be bounded? We have

$$g_1^2(x_1, x_2) = \left(\int_{-\infty}^{x_1} g(s, x_2) \psi(s, x_2) ds \right)^2 \leq \left(\int_{-\infty}^{x_1} g^2(s, x_2) ds \right) \left(\int_{-\infty}^{x_1} \psi^2(s, x_2) ds \right).$$

Therefore, a reasonable condition would be

$$\left(\int_{\mathbb{R}} \psi^2(s, x_2) ds \right)^{\frac{1}{2}} \lesssim \frac{\|g\|_2}{\left(\int_{\mathbb{R}} g^2(s, x_2) ds \right)^{\frac{1}{2}}}.$$

We introduce two useful functions: the width function $x_2 \mapsto w(x_2)$ and the height function $x_1 \mapsto h(x_1)$:

$$h(x_1) = \|g(x_1, \cdot)\|_{L^2(\mathbb{R})}; \quad w(x_2) = \|g(\cdot, x_2)\|_{L^2(\mathbb{R})}.$$

Note that these functions satisfy simple relations

$$\int_{\mathbb{R}} h^2(x_1) dx_1 = \|g\|_{L_2}^2; \quad \int_{\mathbb{R}} w^2(x_2) dx_2 = \|g\|_{L_2}^2.$$

Recall that we want the integral of $(1-\psi)g$ with respect to x_2 to be a bounded function as well. Therefore, we may require the following conditions for ψ :

$$\|\psi(\cdot, x_2)\|_{L^2(\mathbb{R})} \lesssim \frac{\|g\|_2}{w(x_2)}; \quad \|(1-\psi)(x_1, \cdot)\|_{L^2(\mathbb{R})} \lesssim \frac{\|g\|_2}{h(x_1)}.$$

For example, these conditions are satisfied if

$$|\psi(x_1, x_2)| \lesssim \frac{h(x_1)}{w(x_2)}; \quad |1-\psi(x_1, x_2)| \lesssim \frac{w(x_2)}{h(x_1)}.$$

The latter conditions hold true for $\psi(x_1, x_2) = \chi_{\{\frac{h(x_1)}{w(x_2)} \geq 1\}}(x_1, x_2)$.

General case. Let I be a non-empty subset of $[1..d]$ of cardinality n , $0 < n \leq d$. The coordinates whose numbers do not belong to I form a $(d-n)$ -dimensional variable x_I . The complement to I is denoted by $\bar{I}$. Thus, $x = (x_I, x_{\bar{I}})$. Let $G \in L_1(\mathbb{R}^d)$ be an arbitrary function. We will be using the notation

$$\int_I G(x_I) = \int_{\mathbb{R}^n} G(x_I, x_{\bar{I}}) dx_{\bar{I}}.$$

According to the Fubini theorem, this function is well defined for almost all $x_I \in \mathbb{R}^{d-n}$ and is summable as a function on $\mathbb{R}^{d-n}$. Let us formulate a statement that slightly strengthens Proposition 1; this statement is more convenient for induction.

Let $g \in L_d$. There exists a splitting of $\mathbb{R}^d$ into disjoint sets A_j , $j \in [1..d]$, such that

$$\left\| \int_{\{j\}} |g \chi_{A_j}| \right\|_{L_\infty(\mathbb{R}^{d-1})} \lesssim \|g\|_{L_d}$$

for all j .

We will be proving this statement by induction with respect to the dimension d . The base case $d = 1$ is trivial (we may use the case $d = 2$ as a base as well). Let us prove the induction step assuming the above assertion holds true in $d-1$ dimensions.

We split $\mathbb{R}^d$ into disjoint sets B_j , $j \in [1..d]$, according to the rule

$$B_j = \left\{ x \mid \forall i \in [1..d] \setminus \{j\} \quad \int_{\overline{\{j\}}} |g|^d(x_j) \leq \int_{\overline{\{i\}}} |g|^d(x_i) \right\}.$$

In other words, for each j we “average” the function $|g|^d$ with respect to all coordinates except the j th one, and after that, we choose the smallest of these averagings. There is a tiny inaccuracy here since we use non-strict inequalities in the definition of the B_j . It is unclear to which set should we assign the points where the inequality turns into equality. In fact, we may choose any of them (of course, this choice should preserve the measurability). We thus obtain the following decomposition for g :

$$g = \sum_{j=1}^d g \chi_{B_j} = \sum_{j=1}^d h_j.$$

If we obtain the desired decomposition for each of the functions h_j , then we will also get a similar decomposition for f .

We fix some j and consider the function $G = |g|^{\frac{d}{d-1}} \chi_{B_j}$ on a $(d-1)$ -dimensional space $\{y \mid y_j = x_j\}$. According to the induction hypothesis, there exist the sets $C_1, C_2, \dots, C_d$ (we enumerate this collection of $d-1$ sets with the numbers from 1 to d excluding j) such that

$$\sum_i \chi_{C_i} = \chi_{\mathbb{R}^{d-1}}, \quad \forall i \neq j \quad \left\| \int_{\{i\}} |G \chi_{C_i}| \right\|_{L_\infty(\mathbb{R}^{d-1})} \lesssim \|G\|_{L^{d-1}}.$$

Let us prove that the choice $A_i = C_i$ provides a similar decomposition for the function g_j (we now “unfreeze” the variable x_j and consider C_i as a subset of $\mathbb{R}^d$ defined “slice-wise”). It suffices to prove the inequality

$$\left\| \int_{\{i\}} |g \chi_{B_j} \chi_{C_i}| \right\|_{L_\infty(\mathbb{R}^{d-1})} \lesssim \|g\|_{L^d}.$$

We use Hölder’s inequality:

$$\int_{\{i\}} |g \chi_{B_j} \chi_{C_i}| \leq \left(\int_{\{i\}} |g|^{\frac{d}{d-1}} \chi_{B_j} \chi_{C_i} \right)^{\frac{d-1}{d}} \left(\int_{\{i\}} \chi_{B_j} \right)^{\frac{1}{d}}. \quad (2)$$

By the construction of B_j , the bound

$$\chi_{B_j}(x) \leq \frac{\int_{\overline{\{i\}}} |g|^d(x_i)}{\int_{\overline{\{j\}}} |g|^d(x_j)}, \quad i \neq j,$$

holds true. Therefore, the expression in the second parenthesis of (2) does not exceed

$$\frac{\|g\|_{L^d}^d}{\int_{\overline{\{j\}}} |g|^d(x_j)}.$$

By the properties of C_i ,

$$\left(\int_{\{i\}} |g|^{\frac{d}{d-1}} \chi_{B_j} \chi_{C_i} \right)^{\frac{d-1}{d}} \leq \left\| \int_{\{i\}} |G \chi_{C_i}| \right\|_{L_\infty(\mathbb{R}^{d-1})}^{\frac{d-1}{d}} \lesssim \|G\|_{L^{\frac{d-1}{d}}}^{\frac{d-1}{d}} \leq \left(\int_{\overline{\{j\}}} |g|^d(x_j) \right)^{\frac{1}{d}},$$

and therefore, (2) implies

$$\int_{\{i\}} |g \chi_{B_j} \chi_{C_i}| \lesssim \frac{\|g\|_{L^d}}{(\int_{\overline{\{j\}}} |g|^d(x_j))^{\frac{1}{d}}} \left(\int_{\overline{\{j\}}} |g|^d(x_j) \right)^{\frac{1}{d}} = \|g\|_{L^d}.$$

□

References

- [1] J. Bourgain, H. Brezis, *On the equation $\operatorname{div} Y = f$ and application to control of phases*, Journ. of AMS, **16**:2 (2002), 393–426.